

# O'Bayes relMaged: probabilistic approximations of possibilistic inferential models

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*Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information. —Efron*

- No prior distribution represents “absence of prior information”
- So, goal is to effectively/cleverly hack Bayes’s theorem
- If we’re looking for a hack, then
  - likelihood-times-prior arguably isn’t essential
  - other constructions are fair game
- My relMaged proposal: *probabilistic approximation of a possibilistic inferential model* (IM)

*Inferential models are one of the original statistical innovations of the 2010s...* —Cui & Hannig

- A possibilistic IM is a mapping from data, model, etc to a possibility distribution for inference
- Mathematical form of IM output: *imprecise probability*
- Imprecision is warranted, but statisticians are unfamiliar
- *Question:* Can we approximate the IM's possibilistic output by a precise probability distribution?
- If the approximation preserves (most of) the IM's reliability, then it's candidate solution to Efron's problem

- Outline:
  - background: possibilistic IMs & properties
  - probabilistic approximations of possibilistic IMs
  - properties
  - computation
  - examples
  - conclusion, open questions, etc
- Setup & notation:
  - data  $Y \in \mathbb{Y}$ , observed value  $y$
  - model  $\{P_\theta : \theta \in \mathbb{T}\}$ , uncertain true value is  $\Theta$
  - vacuous prior info about  $\Theta$
- Goal is inference/UQ about  $\Theta$ , given  $Y = y$

## Definition — *inferential model* (IM).

A model for quantifying uncertainty about  $\Theta$  given  $y$ , model, etc:

- to every  $H \subseteq \mathbb{T}$ , assign a pair of numbers

$$\underline{\Pi}_y(H) \quad \text{and} \quad \overline{\Pi}_y(H)$$

- $y$ -dependent measures of support for and plausibility of  $H$
- Distinct lower and upper prob's implies *imprecision*
- Precise case corresponds to taking  $\underline{\Pi}_y = \overline{\Pi}_y$
- But imprecision isn't a shortcoming, flaw, or an unnecessary abstraction — it's actually necessary for reliability!

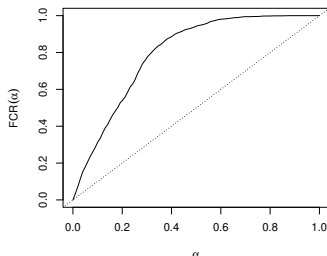
*It is unacceptable if a procedure... of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions. —Reid & Cox*

- Let  $Q_Y$  be a data-dependent probability
- Problematic situation:
  - for some false hypotheses  $H \not\in \Theta$
  - the rv  $Y \mapsto Q_Y(H)$  tends to be large under  $P_\Theta$
- Define  $\text{FCR}_Q(\alpha, H) = \sup_{\theta \notin H} P_\theta\{Q_Y(H) > 1 - \alpha\}$

False confidence theorem (Balch et al, arXiv:1706.08565).

Let  $Q_Y$  be *any* data-dependent probability for  $\Theta$ . For any  $(\alpha, \beta) \in (0, 1)$ , there exists  $H \subset \mathbb{T}$  such that  $\text{FCR}_Q(\alpha, H) > \beta$

- Simple linear regression:  $(Y_i | x_i) \stackrel{\text{ind}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$
- Covariates are fixed, model parameter  $\theta = (\beta_0, \beta_1, \sigma^2)$
- $Q_y$  = Bayes posterior w/ diffuse normal–gamma prior
- Simulation setup:
  - $H = \{(\beta_0, \beta_1, \sigma^2) : -\beta_0/\beta_1 > -1\}$  <sup>2</sup>
  - data generated with  $\Theta = (0.3, 0.1, 1)$ , so  $H$  is false



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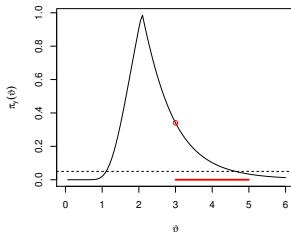
<sup>2</sup>“Non-linearity” in  $H$  causes false confidence... (M., arXiv:2404.16228)

- FCT: reliability requires an imprecise IM
- *Possibility theory* is the simplest imprecise probability
- Similar to probability theory
  - determined by a “density function”
  - but with a different calculus — optimization vs integration
- Specifically:
  - possibility contour  $\theta \mapsto \pi_y(\theta)$  with  $\sup_{\theta} \pi_y(\theta) = 1$
  - plausibility/upper probability

$$\overline{\Pi}_y(H) = \sup_{\theta \in H} \pi_y(\theta), \quad H \subseteq \mathbb{T}$$

- support/lower probability  $\underline{\Pi}_y(H) = 1 - \overline{\Pi}_y(H^c)$

- Illustration of  $\theta \mapsto \pi_y(\theta)$
- Hypothesis  $H = [3, 5]$
- $\bar{\Pi}_y(H)$  via optimization



- $\bar{\Pi}_y$  has an associated *credal set*,  $\mathcal{C}(\bar{\Pi}_y)$ , contents are precise probabilities with  $Q_y(\cdot) \leq \bar{\Pi}_y(\cdot)$
- For possibility measures, there's a nice characterization:

$$Q_y \in \mathcal{C}(\bar{\Pi}_y) \iff Q_y\{\pi_y(\Theta) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1]$$

- Special case of a general construction<sup>3</sup>
- Workhorse: *probability-to-possibility transform*
- For the present “no-prior” case:
  - let  $R(y, \theta) = L_Y(\theta) / \sup_{\vartheta} L_Y(\vartheta)$  be relative likelihood
  - define a possibility contour

$$\pi_Y(\theta) = P_{\theta}\{R(Y, \theta) \leq R(y, \theta)\}, \quad \theta \in \mathbb{T}$$

- then  $\bar{\Pi}_Y(H) = \sup_{\theta \in H} \pi_Y(\theta)$ , etc.
- Not unfamiliar — p-values!<sup>4</sup>
- However:
  - not significance testing, fully conditional *possibilistic UQ*
  - Fisher: *p-value does not justify any exact probability statement*

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<sup>3</sup>M., arXiv:2211.14567

<sup>4</sup>M., arXiv:1203.6665 and arXiv:1511.06733

## Strong reliability theorem.

$$\sup_{\theta} P_{\theta}\{\pi_Y(\theta) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1]$$

- Consequently, the possibilistic IM controls false confidence rate, i.e.,  $\text{FCR}_{\underline{\Pi}}(\alpha, H) \leq \alpha$  for all  $H$
- “Strong” because this isn’t just significance testing<sup>5</sup>

$$\sup_{\theta} P_{\theta}\{\underbrace{\overline{\Pi}_Y(H) \leq \alpha \text{ for some } H \ni \theta}_{\text{uniformity in } H}\} \leq \alpha, \quad \alpha \in [0, 1]$$

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<sup>5</sup>Cella and M., arXiv:2304.05740

## Corollary.

- 1 the test “reject  $H$  iff  $\overline{\Pi}_Y(H) \leq \alpha$ ” satisfies

$$\sup_{\theta \in H} P_{\theta} \{ \overline{\Pi}_Y(H) \leq \alpha \} \leq \alpha, \quad \alpha \in [0, 1], \quad H \subseteq \mathbb{T}$$

- 2 the region  $C_{\alpha}(y) = \{ \theta : \pi_y(\theta) > \alpha \}$  satisfies

$$\sup_{\theta} P_{\theta} \{ C_{\alpha}(Y) \not\ni \theta \} \leq \alpha, \quad \alpha \in [0, 1]$$

## Corollary.

Elements of IM's credal set are *confidence distributions*, i.e.,

$$Q_Y \in \mathcal{C}(\overline{\Pi}_Y) \iff Q_Y \{ C_{\alpha}(y) \} \geq 1 - \alpha$$

- Possibilistic IMs are “asymptotically Gaussian”<sup>6</sup>
- Analogous to Bayesians’ *Bernstein–von Mises theorem*
- Implications:
  - IMs are “efficient” in the classical sense
  - level sets  $C_\alpha(y)$  are approximately elliptical...

## Possibilistic Bernstein–von Mises theorem.

Simplest-to-state version: under the usual regularity conditions,

$$\sup_{z \in \text{compact}} |\pi_{Y^n}(\hat{\theta}_{Y^n} + J_{Y^n}^{-1/2} z) - \gamma(z)| \rightarrow 0 \quad \text{in } P_\Theta\text{-probability,}$$

where  $\gamma(z) = P(\|Z\|^2 > \|z\|^2)$  is a standard Gaussian possibility contour

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<sup>6</sup>M. and Williams, [arXiv:2412.15243](https://arxiv.org/abs/2412.15243)

# Summary so far

- Probabilistic UQ can't guarantee reliability
- Possibilistic UQ can — difference is in the calculus
- The construction above does it efficiently<sup>7</sup>
- *More than significance testing!*

$$\left\{ \begin{array}{l} \text{marginalization}^* \\ \text{decision-making} \\ \text{prediction} \\ \text{model assessment} \\ \dots \end{array} \right. \quad \longleftarrow \text{Choquet integration wrt } \overline{\Pi}_y$$

- But, possibility theory, etc is unfamiliar...

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<sup>7</sup>Good statistical and possibility-theoretic reasons for using relative likelihood ranking, but other considerations (\*) might suggest a different ranking; regardless of the ranking choice, strong reliability holds

- Approximate  $\overline{\Pi}_y$  by an element in  $\mathcal{C}(\overline{\Pi}_y)$ ?
- Recall:  $Q_y \in \mathcal{C}(\overline{\Pi}_y)$  iff  $Q_y\{C_\alpha(y)\} \geq 1 - \alpha$
- Roughly, an *inner probabilistic approximation* of  $\overline{\Pi}_y$  is a “most diffuse” member of  $\mathcal{C}(\overline{\Pi}_y)$
- Inner probabilistic approximation  $Q_y^*$  satisfies

$$Q_y^*\{C_\alpha(y)\} = 1 - \alpha, \quad \alpha \in [0, 1]$$

- Usually exists,<sup>8</sup> but not unique
- $Q_y^*$  is exactly *probability-matching*
- Not practically useful if we don't know what  $Q_y^*$  looks like...

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<sup>8</sup>As stated, existence fails, e.g., if  $\pi_y$  is bounded away from 0

Theorem (M., arXiv:2501.10585).

Each member  $Q_y$  of  $\mathcal{C}(\overline{\Pi}_y)$  can be expressed as

$$Q_y(\cdot) = \int_0^1 K_y^\beta(\cdot) M(d\beta)$$

- $M$  is a probability stochastically no smaller than  $\text{Unif}(0, 1)$
- $K_y^\beta$  is a probability fully supported on  $C_\beta(y)$  for each  $\beta \in [0, 1]$

Corollary.

Inner probabilistic approximations  $Q_y^*$  correspond to

- $M = \text{Unif}(0, 1)$
- $K_y^\beta$  fully supported on  $\partial C_\beta(y)$  for each  $\beta \in [0, 1]$

- General proposal: approximate the possibilistic IM output by the probability distribution  $Q_y^*$  with
  - $M = \text{Unif}(0, 1)$
  - $K_y^\beta = \text{Unif}\{\partial C_\beta(y)\}$  for each  $\beta \in [0, 1]$
- Conceptually simple: uniform mixture of uniforms<sup>9</sup>
- Computationally non-trivial — more on this later
- Is this even a reasonable alternative to default-prior Bayes?
- Recall:  $Q_y^*$  is probability-matching
- Any other relevant properties?

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<sup>9</sup>Uniform kernel isn't necessary; motivated by indifference principles, etc

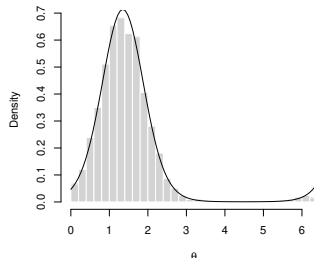
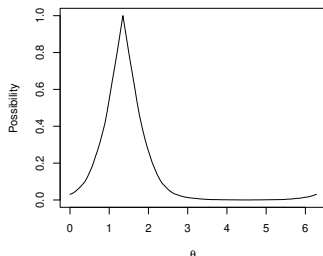
- Consider an invariant statistical model, e.g., location–scale
- Group structure suggests a default, right Haar prior
- Bayes and fiducial solutions agree in such cases, and they're also inner probabilistic approximations of  $\bar{\Pi}_y$

Theorem (M., [arXiv:2503.19748](#)).

Under the above setup, the right Haar prior Bayes posterior distribution is an inner probabilistic approximation of  $\bar{\Pi}_y$

# Probabilistic approximation, cont.

- Example: von Mises distribution for directional data
- Invariant with respect to rotations
- Plot below shows
  - possibilistic IM contour function  $\pi_y$
  - samples from the inner probabilistic approximation
  - right Haar prior Bayes posterior density function



- Connection to Bayes suggests some other nice properties
- Should at least expect desirable asymptotic properties
- Possibilistic BvM for  $\bar{\Pi}_y$  implies a standard BvM for  $Q_y^*$
- Implies  $Q_y^*$  is asymptotically efficient, in addition to being exactly probability-matching

Theorem (M., arXiv:2503.19748).

Simplest-to-state: Under usual regularity conditions,  $Q_{Y^n}^*$  asymptotically merges with  $N(\hat{\theta}_{Y^n}, J_{Y^n}^{-1})$ , with probability tending to 1

- Sketch some details in the scalar-parameter case
- $C_\alpha(y) = [a_\alpha(y), b_\alpha(y)]$ , with  $a_\alpha(y) < \hat{\theta}_y < b_\alpha(y)$
- Note:  $\{a_\alpha(y) \geq \theta\}$  or  $\{b_\alpha(y) \leq \theta\} \iff \pi_y(\theta) \leq \alpha$
- CDF of  $Q_y^*$ :

$$\begin{aligned} Q_y^*((-\infty, \theta]) &= \int_0^1 \left[ \frac{1}{2} 1\{a_\alpha(y) \leq \theta\} + \frac{1}{2} 1\{b_\alpha(y) \leq \theta\} \right] d\alpha \\ &= \begin{cases} \frac{1}{2} \pi_y(\theta) & \text{if } \theta \leq \hat{\theta}_y \\ 1 - \frac{1}{2} \pi_y(\theta) & \text{if } \theta > \hat{\theta}_y. \end{cases} \end{aligned}$$

- Local coordinates,  $\theta = \hat{\theta}_y + J_y^{-1/2}z$ , e.g.,

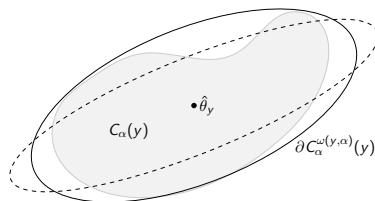
$$\frac{1}{2} \pi_y(\hat{\theta}_y + J_y^{-1/2}z) \rightarrow \frac{1}{2} P(Z^2 \geq z^2) = P(Z \leq z), \quad z \leq 0$$

- Mathematically,  $Q_y^*$  is simple, i.e., a uniform mixture
- Computation is nontrivial because:
  - the kernels are supported on the boundary,  $\partial C_\alpha(y)$
  - and that boundary can't be easily found
- But we're not flying blind — the possibilistic BvM implies that the  $\partial C_\alpha(y)$ 's are *approximately elliptical*
- If  $\partial C_\alpha(y)$  is exactly elliptical, then sampling there is easy<sup>10</sup>
- Idea: approximate  $\partial C_\alpha(y)$  by a  $(y, \alpha)$ -dependent ellipsoid

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<sup>10</sup>Linear transformation of a uniform draw on unit ball

- Define  $C_\alpha^\omega(y) = \{\theta : (\theta - \hat{\theta}_y)^\top J_y(\omega)^{-1}(\theta - \hat{\theta}_y) \leq \chi_{1-\alpha}^2\}$
- $J_y(\omega)$  is an eigenvalue-adjusted observed Fisher info
- Choose  $\omega(y, \alpha)$  such that  $C_\alpha(y) \subseteq C_\alpha^{\omega(y, \alpha)}(y)$ , snugly
- How to get  $\omega(y, \alpha)$  is discussed in arXiv:2501.10585<sup>11</sup>
- Containment makes approx'n conservative  $\implies$  valid



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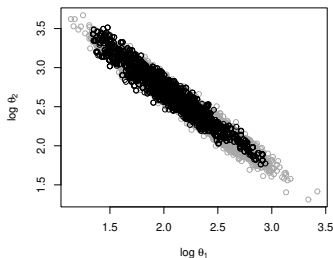
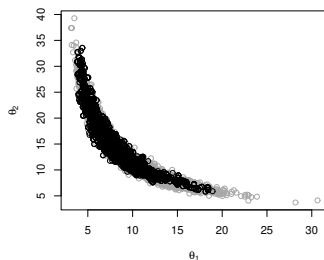
<sup>11</sup>See, also, Cella and M., arXiv:2404.19224

## Approximate Monte Carlo sampling from $Q_y^*$

- 1 Pre-compute  $\omega(y, \alpha_g)$  on a grid  $\alpha_1, \dots, \alpha_G$
  - 2 For each  $m = 1, \dots, M$ , do the following
    - draw  $A \sim \text{Unif}(0, 1)$
    - set  $\omega(A)$  equal to  $\omega(y, \alpha_g)$  for  $\alpha_g$  closest to  $A$
    - draw  $(\Theta_m \mid y, A) \sim \text{Unif}\{C_A^{\omega(A)}(y)\}$
  - 3 Do whatever you usually do with a posterior sample  $\Theta_1, \dots, \Theta_M$
- Akin to familiar Monte Carlo — intentionally sample from an incorrect dist'n because the correct one is complicated
  - Not just a Gaussian approx'n since  $\omega$  depends on  $\alpha$
  - Elliptical approximation is often very accurate; transforming the parameter first often improves accuracy

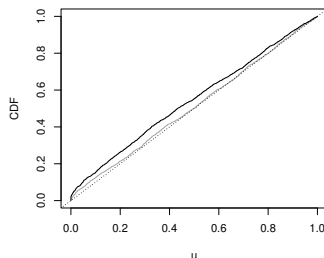
# Example: gamma model

- Real data example,  $n = 20$
- Plots show Jeffreys prior Bayes (black) inner approx'n (gray)
- So the inner probabilistic approximation is new/different



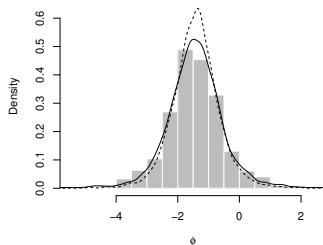
## Example: gamma model, cont.

- Is new/different better?
- Simulation study, settings like real-data example above
- Check for calibration, credible-set coverage
  - Jeffreys prior Bayes (black)
  - inner probabilistic approximation (gray)
- On/below the diagonal means probability matching, i.e., the “right” spread; above diagonal means too concentrated



# Example: Behrens–Fisher problem

- Interest is in the difference  $\Phi$  of the means of two normal populations, unknown variances<sup>12</sup>
- Standard real-data example, plot shows
  - Bayes with right Haar prior (solid)
  - Bayes with Jeffreys prior (dashed)
  - samples from inner probabilistic approximation



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<sup>12</sup>Paper talks about handling nuisance parameters...

## Example: Behrens–Fisher problem, cont.

- Again, is new/different better?
- Simulation study to compare different 90% intervals
- Simple/standard parameter settings
- Unbalanced data:  $n_1 = 3$  and  $n_2 = 20$

| Method                      | Coverage prob | Mean length |
|-----------------------------|---------------|-------------|
| Bayes with Jeffreys prior   | 0.861         | 2.31        |
| Bayes with right Haar prior | 0.929         | 3.26        |
| t.test with Welch df        | 0.881         | 2.79        |
| IM's inner approximation    | 0.904         | 2.71        |

*Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information. —Efron*

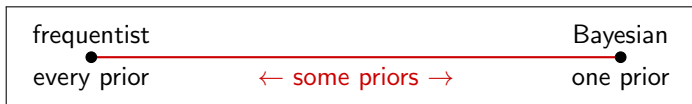
- Maybe the resolution isn't Bayesian or even probabilistic...
- New possibilistic IM framework:
  - likelihood-based
  - familiar and relatively simple
  - achieves strong reliability & efficiency
  - ...
- Today: possibilistic IM can be approximated by a probability distribution that inherits *most of* its desirable properties

- Lots of things that can be done for  $\overline{\Pi}_y$  and/or  $Q_y^*$ ...
- At least partially open theory/methods questions:
  - nuisance parameter elimination
  - decision-theory
  - model assessment/selection
  - prediction
- Interesting & unexpected connections to other things:
  - fiducial and confidence distributions
  - e-values<sup>13</sup>
  - conformal prediction<sup>14</sup>
- Scale up computation for high-dim problems?
- Open applications: *basically all of them!*

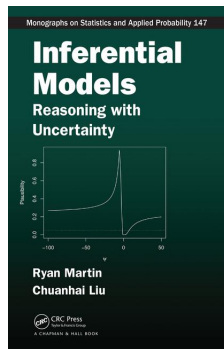
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<sup>13</sup>M., arXiv:2303.17425 and arXiv:2410.01427

<sup>14</sup>Cella and M., arXiv:2001.09225 and arXiv:2112.10234



- I started with “every prior” case  $\longrightarrow$
- But real problems fall in the **middle**
- New  $n$ -part series of working papers:  
*Valid and efficient imprecise-probabilistic  
inference with partial priors*
- Basically  $n = 3$  parts so far
  - 1 arXiv:2203.06703
  - 2 arXiv:2211.14567
  - 3 arXiv:2309.13454





## ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

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*Thanks for your attention!*